CLASS 10 — SCIENCE · NCERT-TYPE UNIT TEST SET 2

Chapters 1–4: **Chemical Reactions and Equations, Acids, Bases and Salts, Metals and Non-metals, Carbon and its Compounds** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. The number of covalent bonds in a molecule of pentane C_5H_{12} is: 1

(a) 14 **(b)** 16 **(c)** 15 **(d)** 17

2. The white precipitate formed on mixing sodium sulphate and barium chloride solutions is: 1

(a) Na_2SO_4 **(b)** $BaSO_4$ **(c)** $NaCl$ **(d)** $BaCl_2$

3. Tooth decay starts when the pH of the mouth falls below: 1

(a) 7 **(b)** 3 **(c)** 5.5 **(d)** 9

4. A metal that reacts with cold water to produce hydrogen but floats because of bubbles of hydrogen: 1

(a) sodium **(b)** potassium **(c)** calcium **(d)** magnesium

5. The cleansing action of soap is due to: 1

(a) formation of micelles **(b)** its smell
(c) reaction with dirt **(d)** its acidity

6. After balancing $H_2 + Cl_2 \longrightarrow HCl$, the coefficient of HCl is: 1

(a) 3 **(b)** 4 **(c)** 2 **(d)** 1

7. Phenolphthalein turns pink in: 1

(a) basic solution **(b)** any solution **(c)** neutral solution **(d)** acidic solution

8. In the electrolytic refining of copper, the anode is made of: 1

(a) impure copper **(b)** zinc **(c)** pure copper **(d)** graphite

9. Carbon forms covalent bonds because it has: 1

(a) six valence electrons **(b)** eight valence electrons
(c) two valence electrons **(d)** four valence electrons

10. A solution of substance X is used for whitewashing. X is: 1

(a) Calcium carbonate **(b)** Sodium hydroxide
(c) Calcium sulphate **(d)** Calcium oxide (quick lime)

11. 10 mL of NaOH solution is completely neutralised by 8 mL of HCl. If 20 mL of the same NaOH is taken, the HCl needed is: 1

(a) 16 mL **(b)** 12 mL **(c)** 8 mL **(d)** 4 mL

12. An element reacts with oxygen to give a compound with a high melting point, soluble in water. The element is likely to be: 1

(a) iron **(b)** calcium **(c)** silicon **(d)** carbon

13. Which of the following is an unsaturated hydrocarbon? 1

(a) C_2H_6 **(b)** C_2H_4 **(c)** CH_4 **(d)** C_3H_8

14. Oxidation is the: 1

(a) loss of oxygen or gain of hydrogen **(b)** gain of oxygen or loss of hydrogen
(c) gain of both oxygen and hydrogen **(d)** loss of both oxygen and hydrogen

15. The acid present in tamarind is: 1

(a) lactic acid **(b)** citric acid **(c)** tartaric acid **(d)** oxalic acid

SECTION — B (Very Short Answer Type) — 5 × 2 = 10

Answer in about 20–30 words each.

16. Why does an aqueous solution of an acid conduct electricity? 2

17. What is aqua regia? Why is it important? 2

18. What is a functional group? Name the one in ethanol. 2

19. What is observed when zinc granules are added to copper sulphate solution? Name the type of reaction. 2

20. Write the chemical name and formula of bleaching powder. 2

SECTION — C (Short Answer Type) — 5 × 3 = 15

Answer in about 40–60 words each.

21. Explain the oxidation of ethanol to ethanoic acid. Name two oxidising agents. 3

22. A solution of substance X is used for whitewashing. Name X and write its reaction with water. 3

23. Write what happens when HCl dissolves in water. What is hydronium ion? 3

24. Differentiate between a metal and a non-metal in respect of 3
(i) conduction
(ii) oxides.

OR

Differentiate between saturated and unsaturated hydrocarbons with an example each. 3

25. Give an example of a double displacement reaction other than the one between sodium sulphate and barium chloride. Write its equation and the observation. 3

OR

What happens when a metal carbonate reacts with an acid? Write the reaction for Na_2CO_3 with HCl and the test for the gas. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

26. Define oxidation and reduction. Identify the substance oxidised, reduced, the oxidising agent and the reducing agent in $ZnO + C \longrightarrow Zn + CO$. 5

OR

Describe the chlor-alkali process. Write the equation, name the electrodes where gases are formed and give one use of each product. 5

27. What is corrosion? Describe the conditions for rusting. State four ways to prevent it. 5

OR

What is esterification? Write the reaction, the conditions, the smell and uses of esters. Write the saponification reaction. 5

END OF QUESTION PAPER **Total: 50 Marks**

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